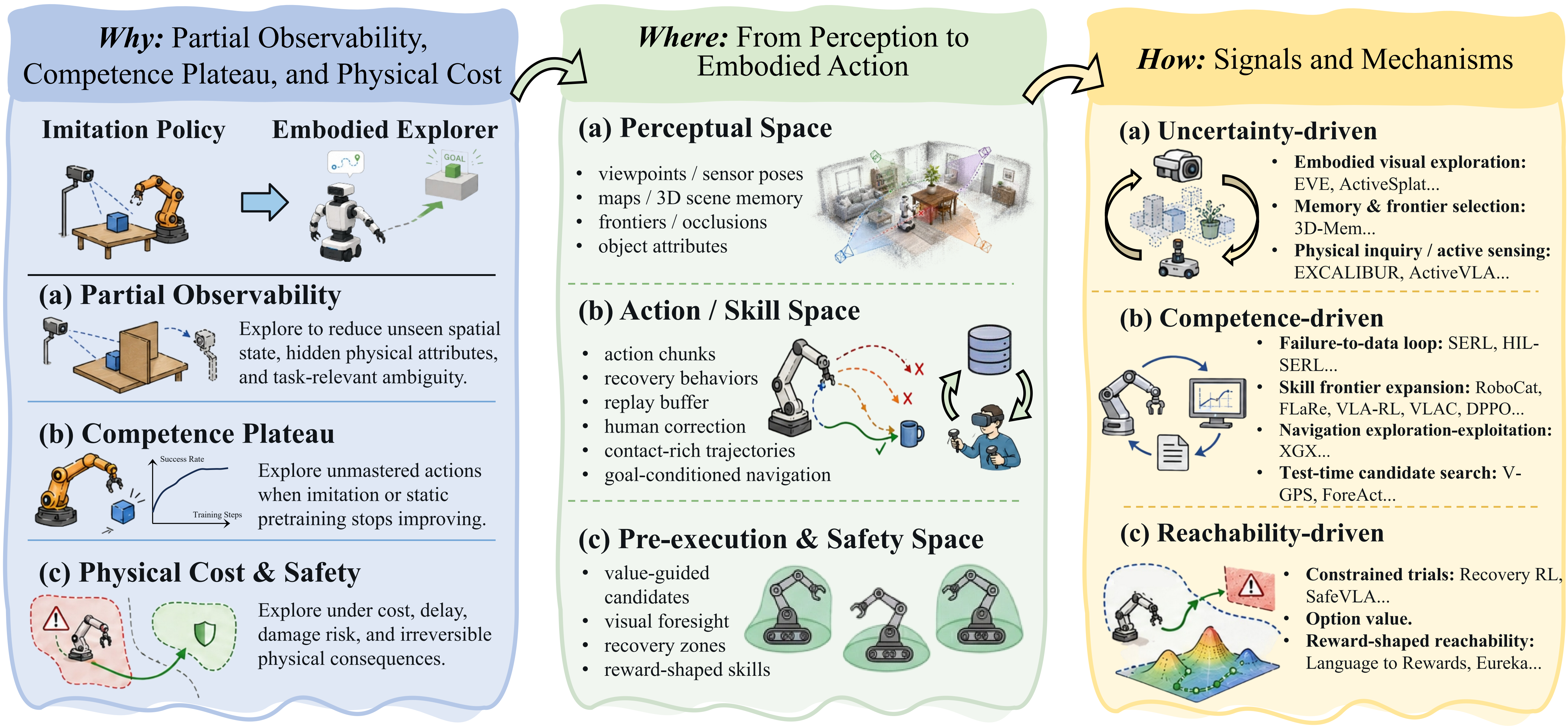

Why: Partial Observability, Competence Plateau, and Physical Cost
Imitation Policy
Embodied Explorer
GOAL
(a) Partial Observability
Explore to reduce unseen spatial state, hidden physical attributes, and task-relevant ambiguity.
(b) Competence Plateau
Success Rate
Training Steps
Explore unmastered actions when imitation or static pretraining stops improving.
(c) Physical Cost & Safety
Explore under cost, delay, damage risk, and irreversible physical consequences.
Where: From Perception to Embodied Action
(a) Perceptual Space
viewpoints / sensor poses
maps / 3D scene memory
frontiers / occlusions
object attributes
(b) Action / Skill Space
action chunks
recovery behaviors
replay buffer
human correction
contact-rich trajectories
goal-conditioned navigation
(c) Pre-execution & Safety Space
value-guided candidates
visual foresight
recovery zones
reward-shaped skills
How: Signals and Mechanisms
(a) Uncertainty-driven
Embodied visual exploration: EVE, ActiveSplat...
Memory & frontier selection: 3D-Mem...
Physical inquiry / active sensing: EXCALIBUR, ActiveVLA...
(b) Competence-driven
Failure-to-data loop: SERL, HIL-SERL...
Skill frontier expansion: RoboCat, FLaRe, VLA-RL, VLAC, DPPO...
Navigation exploration-exploitation: XGX...
Test-time candidate search: V-GPS, ForeAct...
(c) Reachability-driven
Constrained trials: Recovery RL, SafeVLA...
Option value.
Reward-shaped reachability: Language to Rewards, Eureka...